

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Describe the construction and major components of an MHD generator. What is the role of each component in power generation.
- Q.24 What are the different types of solar thermal power plants? Explain any one in detail with a neat diagram
- Q.25 Explain the working principle of geothermal power plants. Discuss the types of geothermal power plants and their respective advantages and disadvantages.

No. of Printed Pages : 4

221564B

Roll No.

6th Sem / Instrumentation & Control

Subject : Energy Management

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Which of the following is a non-renewable source of energy
- a) Solar energy b) Coal
- c) Wind energy. d) Biomass
- Q.2 The energy from falling water is used in
- a) Hydroelectric power plants
- b) Solar plants,
- c) Thermal power plants
- d) Wind farms
- Q.3 Wind energy is converted into
- a) Heat energy b) Electrical energy
- c) Chemical energy d) Sound energy

Q.4 Which renewable energy comes from Earth's internal heat

- a) Solar energy b) Biomass energy
- c) Tidal energy d) Geothermal energy

Q.5 A fuel cell converts _____ energy into electrical energy

- a) Heat b) Mechanical
- c) Chemical d) Nuclear

Q.6 Which of the following is an example of a secondary battery.

- a) Alkaline battery b) Zinc-carbon battery
- c) Lead-acid battery d) Mercury cell

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 The primary source of energy for Earth is:

Q.8 Write one name of non-renewable source of energy

Q.9 Which battery is commonly used in electric vehicles

Q.10 During the charging of a secondary battery, the energy supplied is

Q.11 What is the main purpose of a fuel cell

Q.12 One major drawback of thermal energy is

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Differentiate between renewable and non-renewable sources of energy. Give two examples of each.

Q.14 Describe the working principle of a thermal power plant.

Q.15 What is energy? Explain its importance in daily life

Q.16 Explain how biomass energy is generated. What are its advantages.

Q.17 Describe the role of geothermal energy and where it is commonly used

Q.18 List the advantages and disadvantages of using solar energy

Q.19 Why is energy conservation important? List any three methods of conserving energy

Q.20 What is a Secondary battery? How does it differ from a primary battery.

Q.21 What are the different types of fuel cells? Give examples and typical applications.

Q.22 Differentiate between preliminary, detailed, and investment-grade energy audits